Assignment 1 – Question 2: Birth Weight Analysis

Question 1

a) **central tendency to compare the two approaches.**

Result:

Measures of Central Tendency

The MEANS Procedure

Variable	Mean	Median	Mode
cooperation	75.36	73.00	72.00
competition	77.79	80.00	68.00

Interpretation

The measures of central tendency indicate clear differences between the two teaching approaches. The mean score under the competitive approach (Mean = 77.79) is higher than that of the cooperative approach (Mean = 75.36). Similarly, the median is higher for the competitive group (80.00) compared to the cooperative group (73.00).

These results suggest that, on average, students in a competitive classroom setting tend to achieve higher academic performance than those in a cooperative setting. The higher median further indicates that at least half of the students in the competitive group scored 80 or above, whereas half of the cooperative group scored 73 or below.

b) **measures of variation to compare the two approaches**

Result:

Measures of Variation

The MEANS Procedure

Variable	Range	Variance	Std Dev
cooperation	15.00	26.71	5.17
competition	28.00	81.10	9.01

Interpretation

The measures of variation show substantial differences in the consistency of student performance between the two groups. The competitive approach exhibits a larger range (28.00), higher variance (81.10), and higher standard deviation (9.01) compared to the cooperative approach, which has a range of 15.00, variance of 26.71, and standard deviation of 5.17.

This indicates that student grades in the competitive classroom are more widely dispersed, reflecting greater variability in outcomes. In contrast, the cooperative approach demonstrates more consistent and stable performance, with students' grades clustered closer to the mean.

c) **measures of shape to compare the two approaches**

Result:

Measures of Shape

The MEANS Procedure

Variable	Skewness	Kurtosis
cooperation	0.82	-0.79
competition	-0.40	-1.17

Interpretation

The distribution of grades also differs in terms of shape. The cooperative approach shows positive skewness (0.82), indicating a right-skewed distribution, where most students scored below the mean with a small number of higher scores pulling the average upward. Its negative kurtosis (-0.79) suggests a flatter distribution compared to a normal distribution.

In contrast, the competitive approach exhibits negative skewness (-0.40), implying a slight left-skewed distribution, where more students achieved higher scores, with fewer low-performing outliers. The more pronounced negative kurtosis (-1.17) indicates an even flatter distribution, suggesting a wider spread of scores around the center.

Question 2

Business Deck (Non-Technical Summary)

Key Question:
Does infant birth weight differ based on selected infant and maternal characteristics?

Key Insights:

- Male babies tend to have a higher average birth weight than female babies.
- Babies born to smoking mothers have a substantially lower birth weight compared to those born to non-smoking mothers.
- Babies born to married mothers generally weigh more at birth than those born to unmarried mothers.

Why This Matters:

Lower birth weight is associated with increased health risks. These findings highlight the importance of maternal health behaviors and family stability in improving infant birth outcomes.

Bottom Line:
Maternal smoking and socio-demographic factors are strongly associated with infant birth weight and should be considered in prenatal health strategies.

Technical Deck (Detailed Statistical Analysis)

Research Objective

The objective of this analysis is to investigate whether infant birth weight varies across selected infant and maternal characteristics using independent two-sample t-tests.

Data Description and Audit

The dataset consists of 50,000 observations with no missing values for the birth weight variable. Birth weight has a mean of 3370.76 grams and a standard deviation of 566.39 grams. Binary explanatory variables were selected for analysis, including baby's sex (Boy), maternal smoking status (MomSmoke), and marital status (Married).

Distribution Assessment

Birth weight exhibits moderate negative skewness (-0.76) and positive kurtosis (2.69). Normality tests indicate deviations from a normal distribution; however, given the large sample size, the t-test remains appropriate due to the Central Limit Theorem.

Statistical Methodology

Independent two-sample t-tests were conducted to compare mean birth weight across groups. Statistical significance was evaluated at the 5% significance level ($\alpha = 0.05$).

Results

Birth Weight by Baby Sex:

T-Test: Birth Weight by Baby Sex

The TTEST Procedure

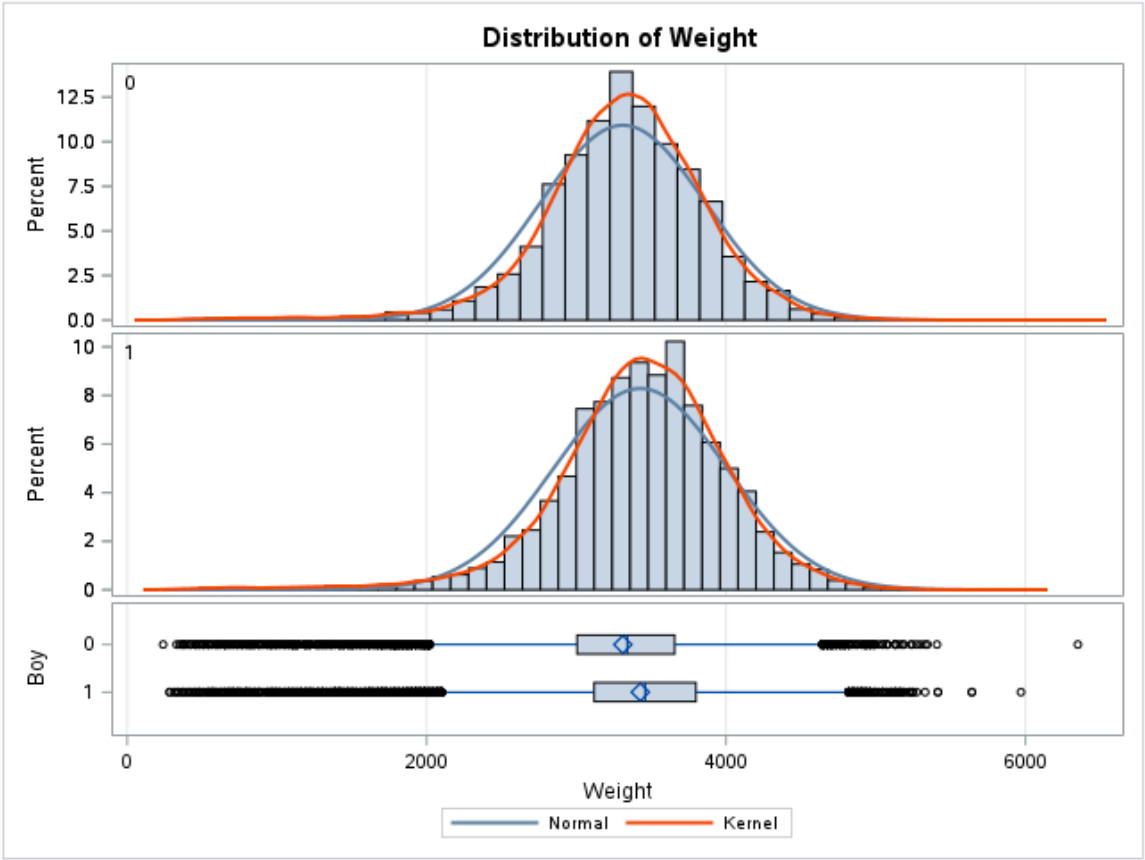
Variable: Weight (Weight)

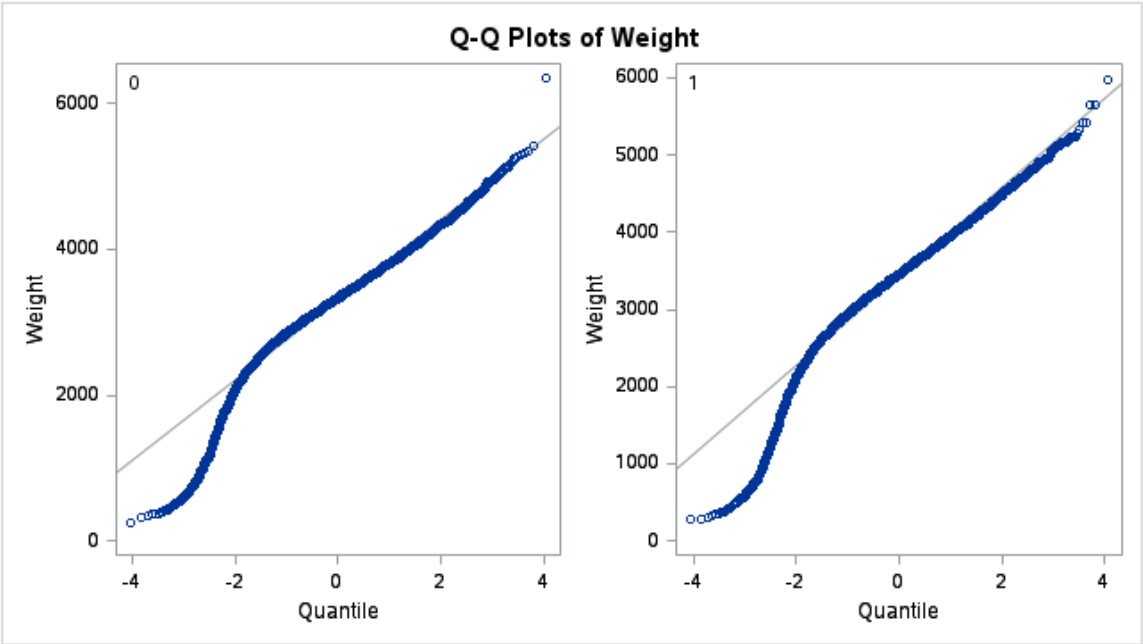
Boy	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		24208	3310.6	547.7	3.5204	240.0	6350.0
1		25792	3427.3	577.7	3.5970	284.0	5970.0
Diff (1-2)	Pooled		-116.7	563.4	5.0416		
Diff (1-2)	Satterthwaite		-116.7		5.0331		

Boy	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3310.6	3303.7	3317.5	547.7	542.9	552.7
1		3427.3	3420.2	3434.3	577.7	572.7	582.7
Diff (1-2)	Pooled	-116.7	-126.6	-106.8	563.4	559.9	566.9
Diff (1-2)	Satterthwaite	-116.7	-126.6	-106.8			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	-23.15	<.0001
Satterthwaite	Unequal	49993	-23.18	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	25791	24207	1.11	<.0001





An independent two-sample t-test was conducted to examine whether infant birth weight differs by baby's sex.

The mean birth weight for male infants was 3427.3 grams *(SD = 577.7)*, whereas the mean birth weight for female infants was 3310.6 grams *(SD = 547.7)*.

The observed mean difference of 116.7 grams indicates that, on average, male babies weigh approximately 117 grams more than female babies at birth. This difference is statistically significant *($t \approx -23.15$, $p < 0.0001$)*, indicating that the observed difference is extremely unlikely to have occurred by random sampling variation alone. The 95% confidence interval for the mean difference [−126.6 g, −106.8 g] does not include zero, providing additional evidence that the difference in mean birth weight between male and female infants is real and not due to chance.

The null hypothesis of equal mean birth weight between male and female infants is rejected. There is strong statistical evidence that infant birth weight differs by baby sex, with male infants having a higher mean birth weight than female infants.

Birth Weight by Maternal Smoking:

T-Test: Birth Weight by Maternal Smoking

The TTEST Procedure

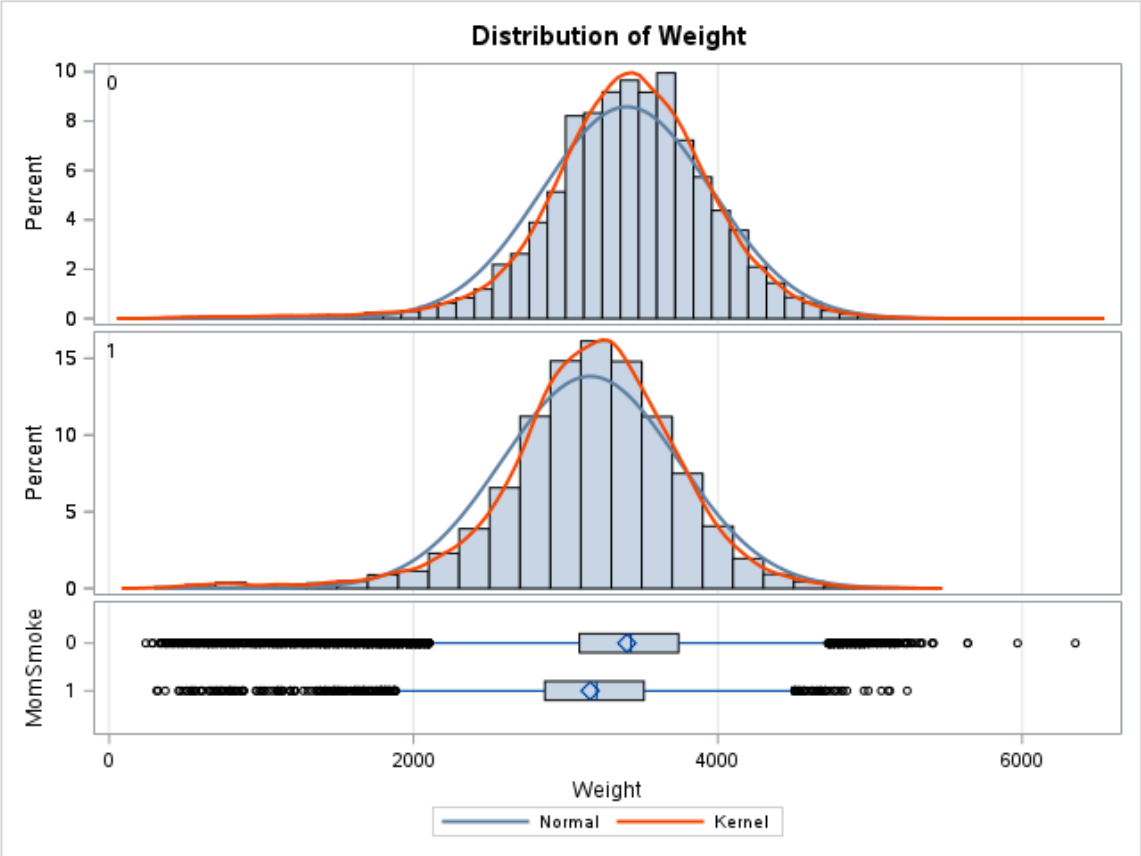
Variable: Weight (Weight)

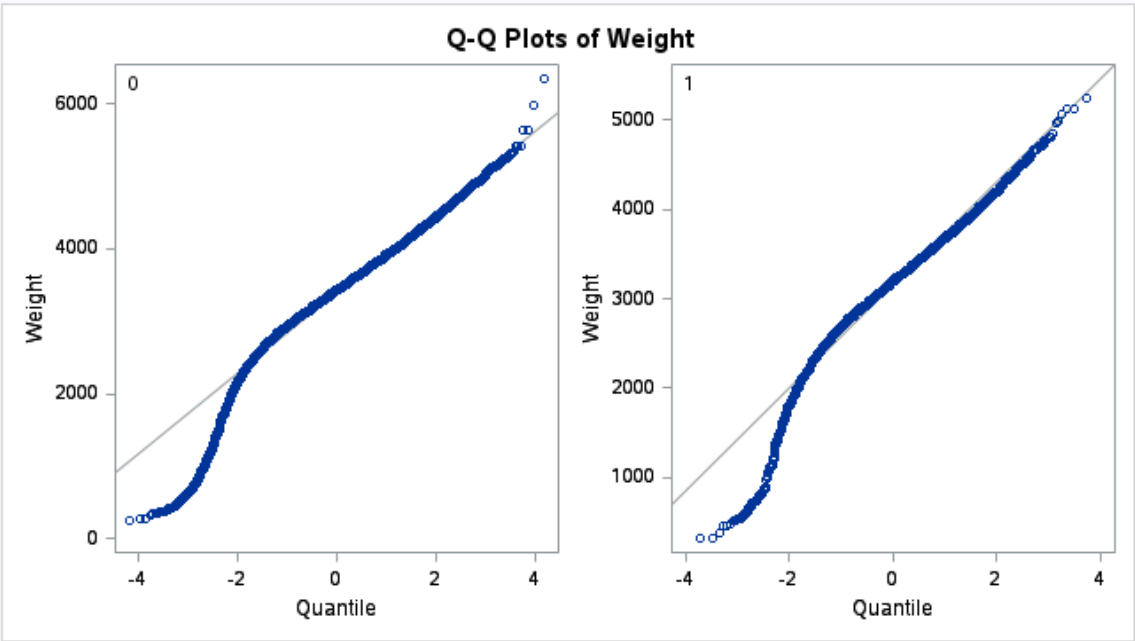
MomSmoke	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		43467	3402.3	558.0	2.6766	240.0	6350.0
1		6533	3160.9	576.8	7.1358	312.0	5245.0
Diff (1-2)	Pooled		241.5	560.5	7.4376		
Diff (1-2)	Satterthwaite		241.5		7.6213		

MomSmoke	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3402.3	3397.1	3407.6	558.0	554.3	561.8
1		3160.9	3146.9	3174.8	576.8	567.0	586.8
Diff (1-2)	Pooled	241.5	226.9	256.0	560.5	557.1	564.0
Diff (1-2)	Satterthwaite	241.5	226.5	256.4			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	32.46	<.0001
Satterthwaite	Unequal	8474.1	31.68	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	6532	43466	1.07	0.0004





An independent two-sample t-test was conducted to examine whether infant birth weight differs by maternal smoking status.

The mean birth weight for infants born to non-smoking mothers was 3402.3 grams (SD = 558.0), whereas the mean birth weight for infants born to smoking mothers was 3160.9 grams (SD = 576.8).The observed mean difference of 241.5 grams indicates that, on average, babies of smoking mothers weigh approximately 242 grams less at birth than babies of non-smoking mothers. This difference is statistically significant ($t \approx 31.68$, $p < 0.0001$), indicating that the observed difference is extremely unlikely to have occurred by random sampling variation alone.The 95% confidence interval for the mean difference [226.5 g, 256.4 g] does not include zero, providing additional evidence that the difference in mean birth weight between infants of smoking and non-smoking mothers is real and not due to chance.

The null hypothesis of equal mean birth weight between infants of smoking mothers is rejected. There is strong statistical evidence that maternal smoking is associated with lower infant birthweight.

Birth Weight by Marital Status:

T-Test: Birth Weight by Marital Status

The TTEST Procedure

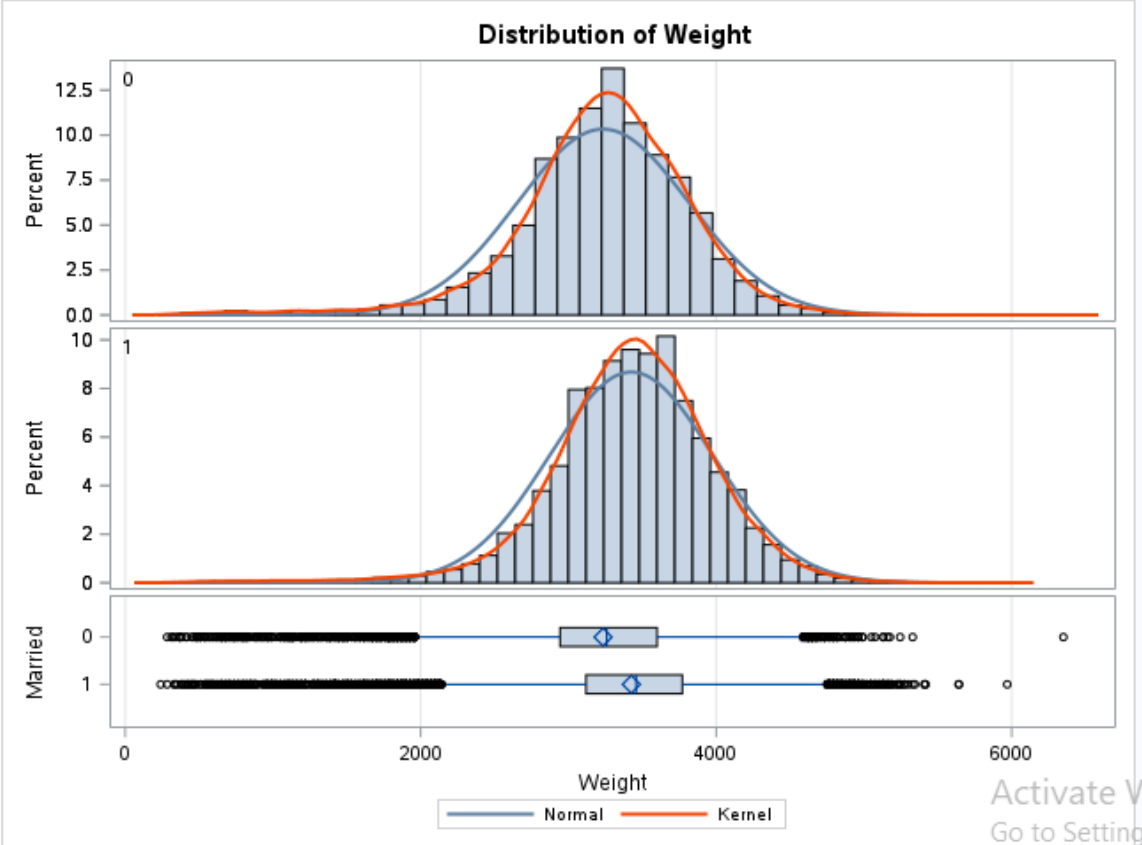
Variable: Weight (Weight)

Married	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		14369	3234.4	579.0	4.8302	284.0	6350.0
1		35631	3425.7	551.8	2.9231	240.0	5970.0
Diff (1-2)	Pooled		-191.3	559.7	5.5315		
Diff (1-2)	Satterthwaite		-191.3		5.6459		

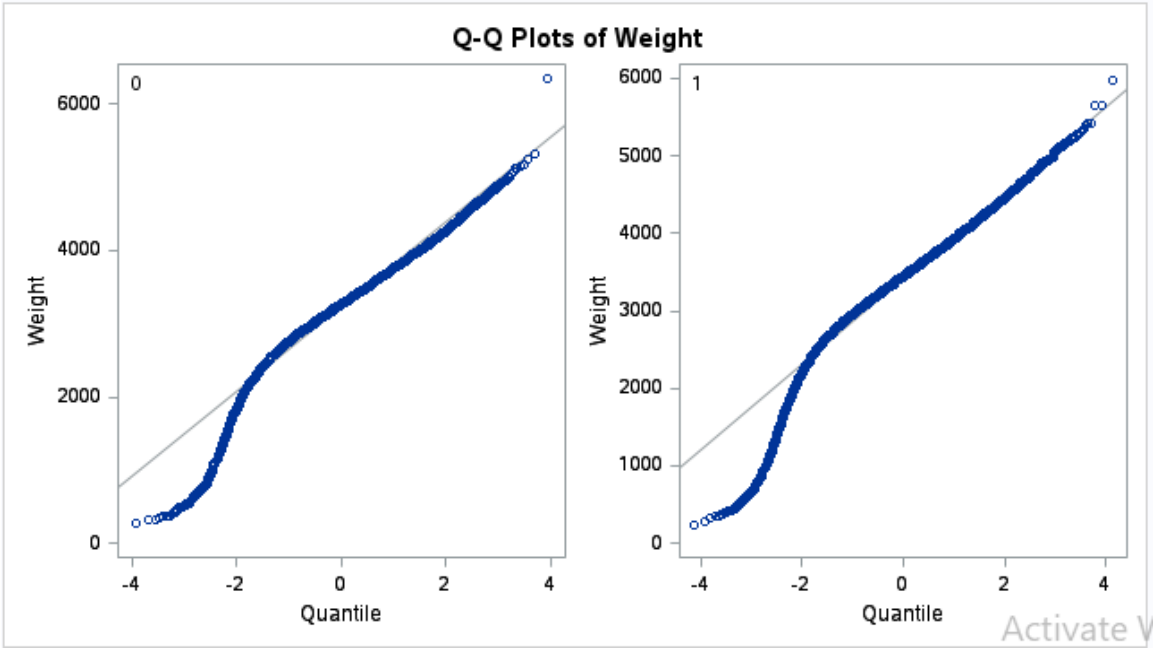
Married	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3234.4	3225.0	3243.9	579.0	572.4	585.8
1		3425.7	3420.0	3431.5	551.8	547.8	555.9
Diff (1-2)	Pooled	-191.3	-202.1	-180.5	559.7	556.3	563.2
Diff (1-2)	Satterthwaite	-191.3	-202.4	-180.2			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	-34.58	<.0001
Satterthwaite	Unequal	25443	-33.88	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	14368	35630	1.10	<.0001



Activate W
Go to Setting



An independent two-sample t-test was conducted to examine whether infant birth weight differs by maternal marital status. The mean birth weight for infants born to unmarried mothers was 3234.4 grams (SD = 579.0), whereas the mean birth weight for infants born to married mothers was 3425.7 grams (SD = 551.8).The observed mean difference of 191.3 grams indicates that, on average, babies born to married mothers weigh approximately 191 grams more at birth than babies born to unmarried mothers. This difference is statistically significant ($t \approx -33.88$, $p < 0.0001$), indicating that the observed difference is extremely unlikely to have occurred by random sampling variation alone.

The 95% confidence interval for the mean difference [−202.4 g, −180.2 g] does not include zero, providing additional evidence that the difference in mean birth weight between infants of married and unmarried mothers is real and not due to chance.

The null hypothesis of equal mean birth weight between infants of married and unmarried mothers is rejected. There is strong statistical evidence that infant birth weight differs by maternal marital status, with infants of married mothers having a higher mean birth weight than infants of unmarried mothers.

Conclusion

The analysis demonstrates that infant birth weight varies significantly with infant sex and maternal characteristics. Maternal smoking shows the strongest association with reduced birth weight. These findings emphasize the importance of prenatal health behaviors and socio-demographic factors in improving birth outcomes.